RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of Artificial Intelligence and Data Science | V Semester B.Tech AIDS

AD23532 – Principles of Data Science

Ex3a: Model Planning and Building - Classification

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Tasks 1 to 5 – Load the dataset, split into Training (80%) and Testing (20%), build the three models

Tasks 1, 2: `CustomerChurn_Preprocessed.csv` *loaded, features and target (Churn_Yes) separated,* `train_test_split(test_size=0.20, random_state=42)`*; models defined*

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    confusion_matrix, accuracy_score,
    precision_score, recall_score, f1_score
)

# Load dataset
df = pd.read_csv("/content/CustomerChurn_Preprocessed.csv")

# Remove customer ID if present
df = df.drop(columns=["customerID"], errors="ignore")

# Convert categorical columns to numeric
# This step is redundant if MF3NEhefhZaE correctly preprocesses and saves the data
# However, to be safe and ensure all columns are numeric before model training,
# we can run it again, but ideally the CSV should already be fully numeric.
# For now, assuming it's needed based on previous context where some were missed.
# If `df` loaded from CSV is already fully numeric, this will have no effect.
df = pd.get_dummies(df, drop_first=True)

# Separate features and target
X = df.drop("Churn_Yes", axis=1)
y = df["Churn_Yes"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

# Models
models = {
```

Tasks 3, 4, 5: Logistic Regression, Decision Tree and Random Forest trained on the training set and used to predict the test set; Task 6 and 7 code

```
    "Decision Tree": DecisionTreeClassifier(random_state=42),
    "Random Forest": RandomForestClassifier(
        n_estimators=100, random_state=42
    )
}

results = {}

# Train and evaluate
for name, model in models.items():

    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)

    cm = confusion_matrix(y_test, y_pred)
    acc = accuracy_score(y_test, y_pred)
    pre = precision_score(y_test, y_pred)
    rec = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)

    results[name] = [acc, pre, rec, f1]

    print("\n", name)
    print("Confusion Matrix:")
    print(cm)
    print("Accuracy :", round(acc, 4))
    print("Precision:", round(pre, 4))
    print("Recall   :", round(rec, 4))
    print("F1-Score :", round(f1, 4))

# Comparison bar chart
result_df = pd.DataFrame(
    results,
    index=["Accuracy", "Precision", "Recall", "F1-Score"]
)

result_df.plot(kind="bar", figsize=(10, 6))
plt.title("Classification Model Comparison")
plt.ylabel("Score")
plt.ylim(0, 1)
plt.xticks(rotation=0)
plt.legend(title="Models")
plt.show()

# Best model
best_model = result_df.mean().idxmax()
print("\nBest Performing Model:", best_model)
```

Task 6 – Model evaluation: Confusion Matrix, Accuracy, Precision, Recall, F1-Score

Logistic Regression

```
 Logistic Regression
Confusion Matrix:
[[934 102]
 [149 224]]
Accuracy : 0.8219
Precision: 0.6871
Recall   : 0.6005
F1-Score : 0.6409
```

Decision Tree

```
 Decision Tree
Confusion Matrix:
[[820 216]
 [200 173]]
Accuracy : 0.7048
Precision: 0.4447
Recall   : 0.4638
F1-Score : 0.4541
```

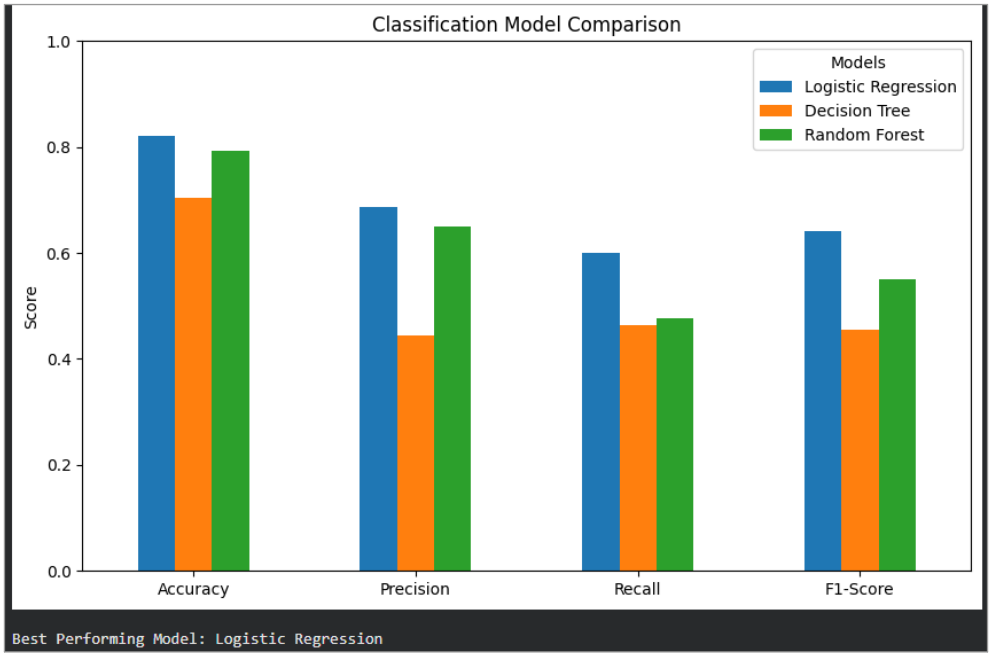
Random Forest

```
 Random Forest
Confusion Matrix:
[[940  96]
 [195 178]]
Accuracy : 0.7935
Precision: 0.6496
Recall   : 0.4772
F1-Score : 0.5502
```

Confusion matrices are in the form [[TN, FP], [FN, TP]]. The test set has 1409 customers; for Logistic Regression, 934 non-churners and 224 churners were predicted correctly.

Task 7 – Comparison of all three models (single bar chart)

Bar chart of Accuracy, Precision, Recall and F1-Score for the three models, and the best performing model printed by the code



Task 8 – Best-performing model

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	**0.8219**	**0.6871**	**0.6005**	**0.6409**
Decision Tree	0.7048	0.4447	0.4638	0.4541
Random Forest	0.7935	0.6496	0.4772	0.5502

Logistic Regression is the best-performing model. It has the highest accuracy (0.8219), precision (0.6871), recall (0.6005) and F1-score (0.6409) of the three models. Random Forest is second (F1-score 0.5502) and Decision Tree is the weakest (F1-score 0.4541).

Task 9 – Business recommendations to reduce customer churn

1. Use the Logistic Regression model to find at-risk customers every month and act before they leave. The model finds about 60% of the customers who actually churn (recall 0.6005, 224 of 373 churners in the test set) and about 69% of the customers it flags really do churn (precision 0.6871). The retention team can call or send offers to the flagged customers first, and the decision threshold can be lowered if the company would rather catch more churners than avoid some false alarms (149 churners were missed).

2. Move month-to-month customers to longer contracts. In the analysis, month-to-month customers churn at 42.7% against 11.3% for one-year and 2.8% for two-year contracts, and churners have much shorter tenure. Discounts, free add-ons or loyalty points for choosing a one- or two-year plan, and extra care during the first months after joining, will reduce churn in the group that leaves most.

3. Fix the pricing and service problems linked to churn. Fiber optic customers churn the most (41.9% against 19.0% for DSL), and churned customers pay higher monthly charges. The company should review fiber plan pricing and service quality, offer bundles with tech support and online security, and encourage automatic bank or card payments, because electronic check is the payment method used by most churned customers (1071).

Result

Thus Logistic Regression, Decision Tree and Random Forest classifiers were trained on the preprocessed Telco customer churn data (80% training, 20% testing) and evaluated using the confusion matrix, accuracy, precision, recall and F1-score. Logistic Regression gave the best results (accuracy 0.8219, F1-score 0.6409), and three recommendations were made to reduce customer churn.